LA HW

Included exercises: 1, 5, 7, 9, 11, 13, 15, 19, 23, 27, 31, 33, 35, 37, 39, 41, 43, 45, 47, 49

Exercises 1-6

In Exercises 1–6, the $\mathcal{B}$-coordinate vector of $\mathbf{v}$ *is given. Find* $\mathbf{v}$ *if*

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \end{bmatrix} \right\}.$$

Original Exercise 1

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

Original Exercise 5

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

Exercises 7-10

In Exercises 7–10, the $\mathcal{B}$-coordinate vector of $\mathbf{v}$ *is given. Find* $\mathbf{v}$ *if*

$$\left\{ \begin{bmatrix} 0 \end{bmatrix} \begin{bmatrix} -1 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix} \right\}$$

Original Exercise 7

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

Original Exercise 9

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 5 \\ -2 \end{bmatrix}$$

Original Exercise 11

(a) Prove that $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$ is a basis for $\mathcal{R}^2$.

(b) Find $[\mathbf{v}]_{\mathcal{B}}$ if $\mathbf{v} = 5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} - 3 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$.

Original Exercise 13

(a) Prove that $\mathcal{B} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix} \right\}$ is a basis for $\mathcal{R}^3$.

(b) Find $[\mathbf{v}]_{\mathcal{B}}$ if $\mathbf{v} = 3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$.

Exercises 15-18

In Exercises 15–18, a vector is given. Find its $\mathcal{B}$-coordinate vector relative to the basis $\mathcal{B}$ in Exercises 1–6.

Original Exercise 15

Exercises 19-22

In Exercises 19–22, a vector is given. Find its $\mathcal{B}$-coordinate vector relative to the basis $\mathcal{B}$ in Exercises 7–10.

Original Exercise 19

$$\begin{bmatrix} 4 \\ 3 \\ 2 \end{bmatrix}$$

Original Exercise 23

Find the unique representation of $\mathbf{u} = \begin{bmatrix} a \\ b \end{bmatrix}$ as a linear combination of

$$\mathbf{b}_1 = \begin{bmatrix} 3 \\ -1 \end{bmatrix} \quad \text{and} \quad \mathbf{b}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Original Exercise 27

Find the unique representation of $\mathbf{u} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ as a linear combination of

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}, \quad \text{and} \quad \mathbf{b}_3 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

Exercises 31-50

In Exercises 31–50, determine whether the statements are true or false.

Original Exercise 31

ments are true or false.

If $\mathcal{S}$ is a generating set for a subspace V of $\mathcal{R}^n$, then every vector in V can be uniquely represented as a linear

Original Exercise 33

vectors in $\mathcal{B}$.

If $\mathcal{E}$ is the standard basis for $\mathcal{R}^n$, then $[\mathbf{v}]_{\mathcal{E}} = \mathbf{v}$ for all $\mathbf{v}$

Original Exercise 35

If $\mathcal{B}$ is a basis for $\mathcal{R}^n$ and B is the matrix whose columns

Original Exercise 37

$\mathbf{v} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_n\mathbf{b}_n$.

If the columns of an $n \times n$ matrix form a basis for $\mathcal{R}^n$,

Original Exercise 39

has a unique solution.

If B is a matrix whose columns are the vectors in a basis $\mathcal{B}$ for $\mathcal{R}^n$, then, for any vector $\mathbf{v}$ in $\mathcal{R}^n$, $[\mathbf{v}]_{\mathcal{B}}$ is a solution

Original Exercise 41

echelon form of $[B \;\; \mathbf{v}]$ is $[I_n \;\; [\mathbf{v}]_{\mathcal{B}}]$.

If $\mathcal{B}$ is any basis for $\mathcal{R}^n$, then $[\mathbf{0}]_{\mathcal{B}} =$

Original Exercise 43

If $\mathbf{v}$ is any vector in $\mathcal{R}^n$, $\mathcal{B}$ is any basis for $\mathcal{R}^n$, and c is

Original Exercise 45

usual x-, y-axes through the angle θ. Then $\begin{bmatrix} x' \\ y' \end{bmatrix} = A_\theta \begin{bmatrix} x \\ y \end{bmatrix}$.

Suppose that the x'-, y'-axes are obtained by rotating the

Original Exercise 47

The graph of an equation of the form $\frac{(x')^2}{a^2} + \frac{(y')^2}{b^2} = 1$ is an ellipse.

Original Exercise 49

The graph of an ellipse with center at the origin can be written in the form $\frac{(x')^2}{a^2} + \frac{(y')^2}{b^2} = 1$ by a suitable